

ZARI.ai Master Final Technical Defense Report

End-to-End Multilingual Agricultural Advisory System: Vision AI, EDL, SCRC, RAG & LLM Engine

1. Executive System Overview

ZARI.ai is an integrated, evidence-grounded AI agricultural advisory system engineered specifically for Pakistani smallholder farming. The system operates on a single unified web server (<http://127.0.0.1:8000>), serving both static web interface components and real-time AI diagnostic microservices.

Master System Specifications

Unified Single Server Port:	http://127.0.0.1:8000
Vision AI Backbone:	2-Stage Hierarchical EfficientNetV2-B2 Suite
Total Model Suite Parameters:	31,109,749 parameters (~31.11 Million Params)
Total Checkpoints Weight Size:	355.73 MB Total File Size on Disk
Input Image Tensor Shape:	RGB (384, 384, 3)
Uncertainty Estimation:	Evidential Deep Learning (Dirichlet Softplus)
Safety Risk Control:	SCRC Threshold $\tau_{SCRC} = 0.3175$ (Dual Safety Gates)
Multilingual Vector RAG:	ChromaDB (208 Chunks) + MiniLM-L12-v2 (384d)
LLM Advisory Generator:	Meta Llama 3.1 8B Instant on Groq LPUs (~750 tok/s)
Offline Fallback Synthesizer:	Deterministic IPM Synthesizer (100% Offline Edge Mode)
Offline End-to-End Latency:	9.40 ms (3.24ms Vision + 5.32ms RAG + 0.86ms Fallback)
Online End-to-End Latency:	389 ms (3.24ms Vision + 5.32ms RAG + 380ms Groq API)

2. Exploratory Data Analysis (EDA): New Dataset & 3-Crop Scope

Below are the verified Exploratory Data Analysis (EDA) figures for the newly ingested dataset (v3) and production 3-crop target dataset:

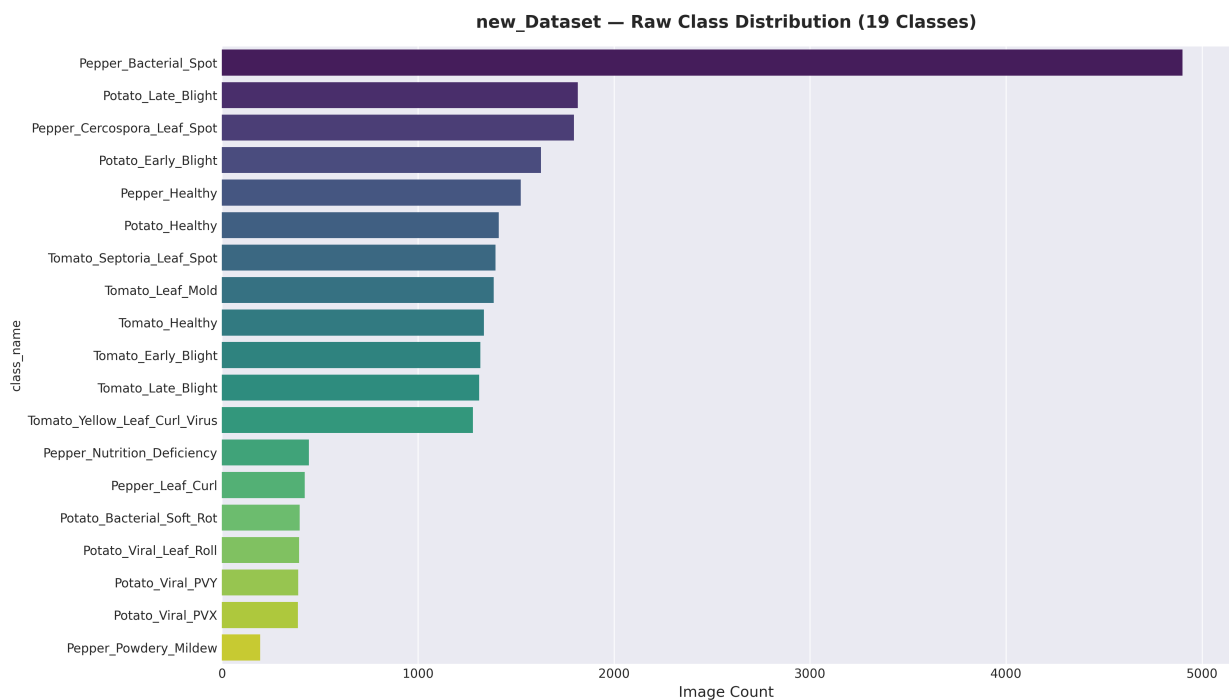


Figure: New Dataset (v3) Class Volume Distribution Across Target Nightshade Diseases

2. Exploratory Data Analysis (Continued)

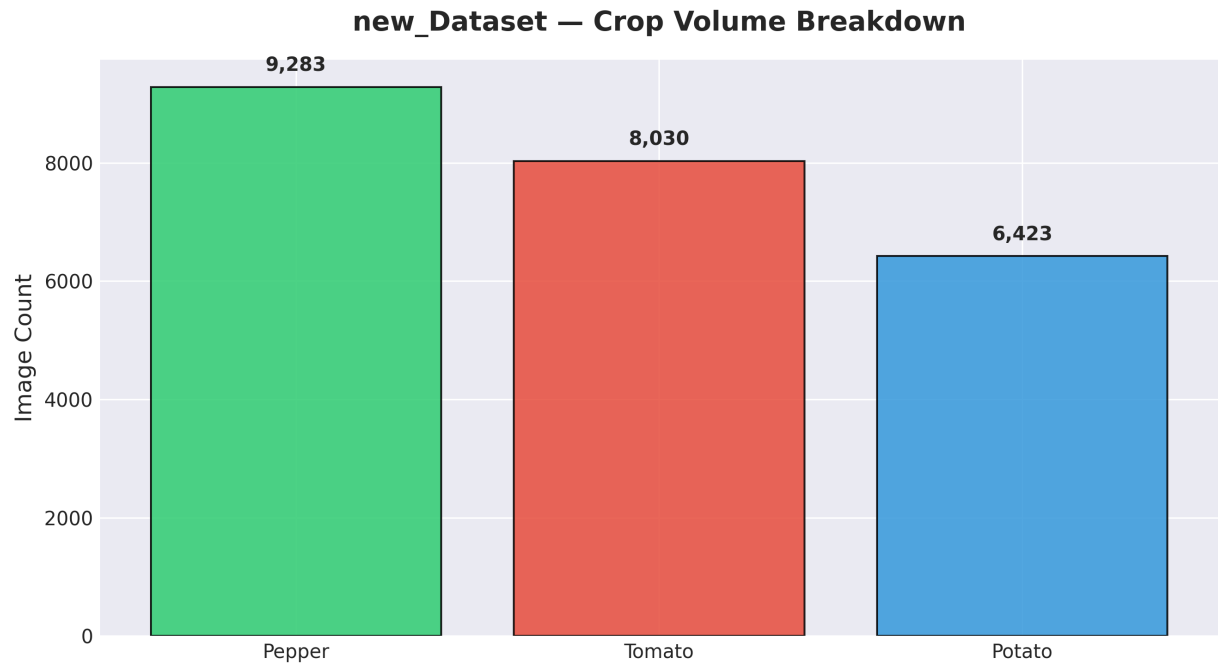


Figure: New Dataset Volume Share by Target Crop Species (Tomato, Potato, Pepper)

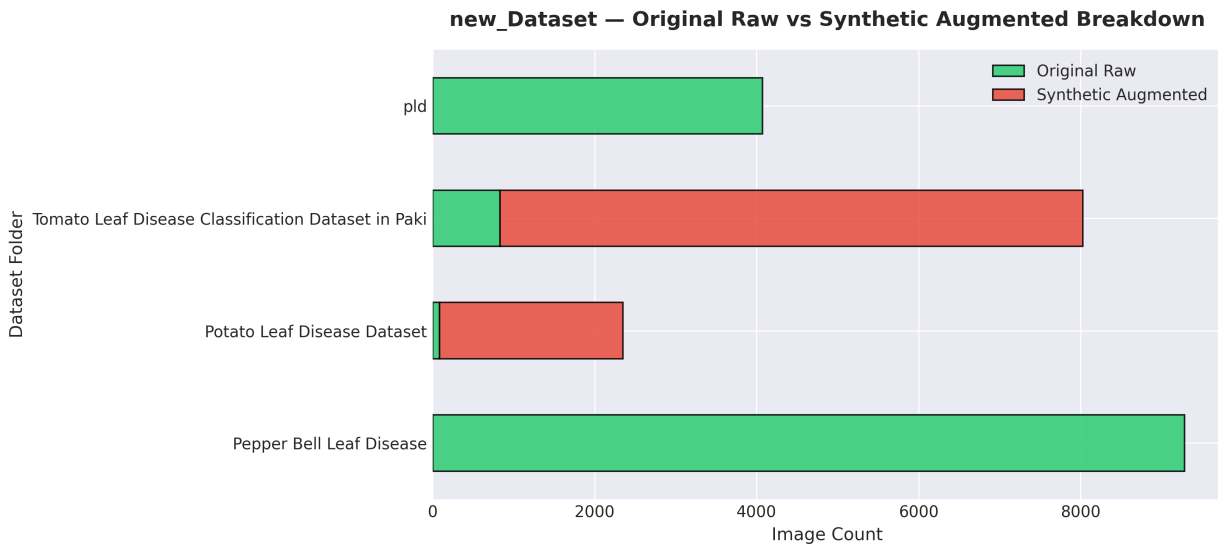


Figure: New Dataset Raw Field Samples vs Synthetic Augmentation Ratios

Production 3-Crop Target Dataset Distributions (31,071 Processed Images)

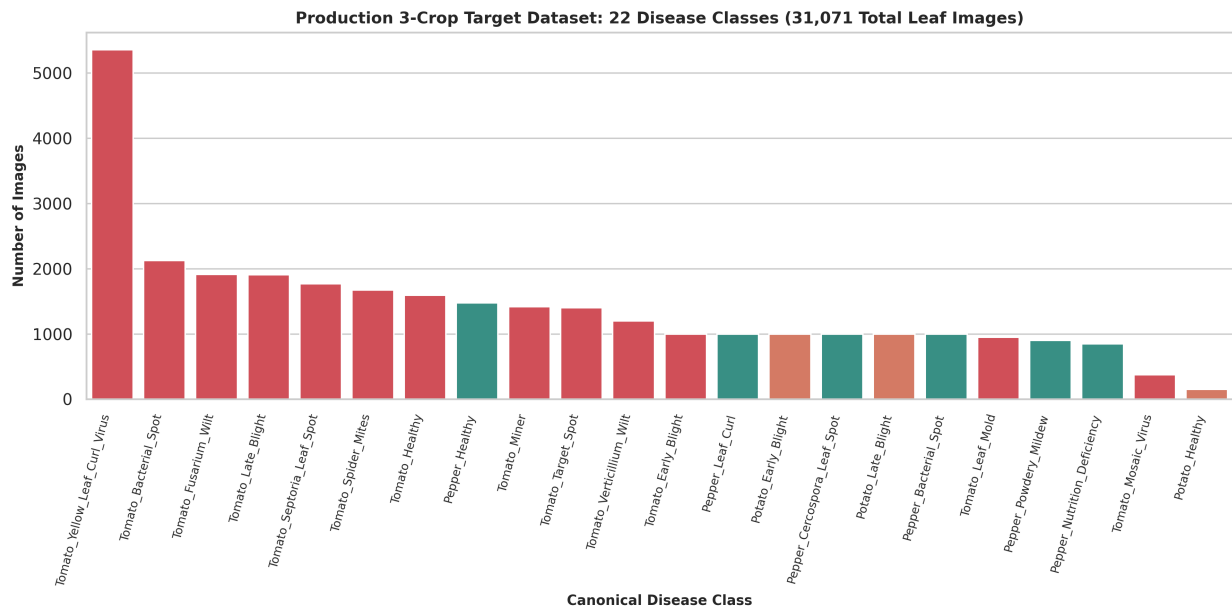


Figure: Production 3-Crop Target Dataset: 22 Disease Classes (31,071 Total Leaf Images)

3-Crop Target Volume Breakdown (31,071 Images)

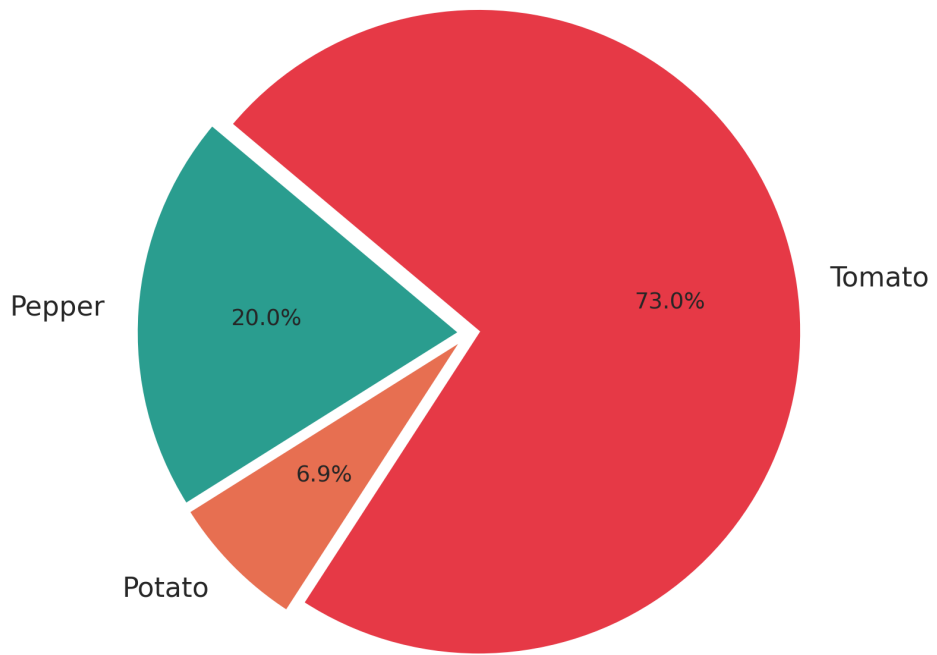


Figure: Production Crop Share (Tomato: 72.8%, Pepper: 20.0%, Potato: 7.2%)

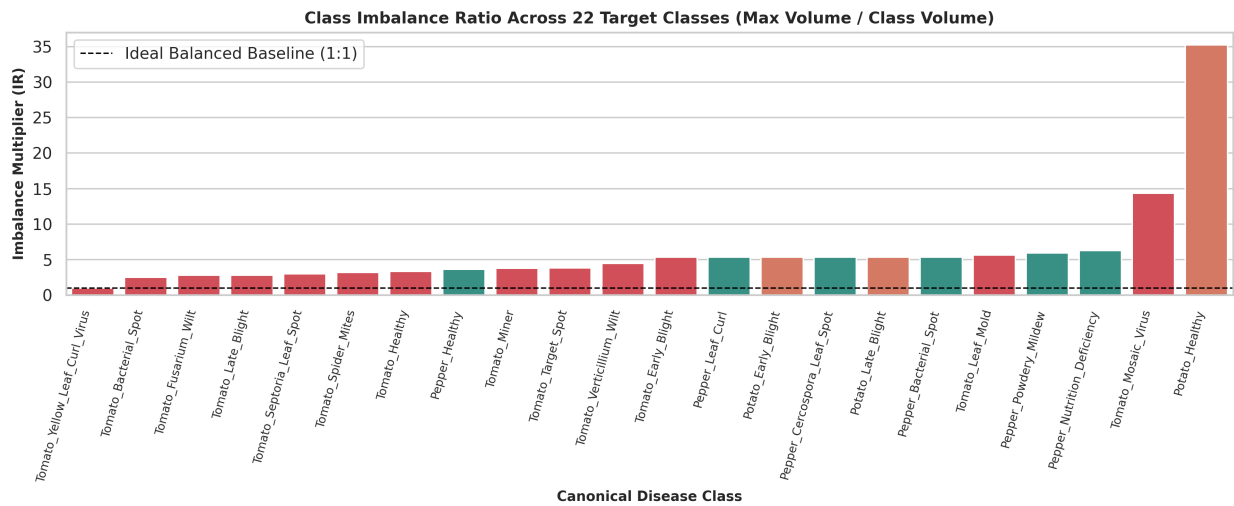


Figure: Class Imbalance Multipliers Across 22 Production Disease Classes

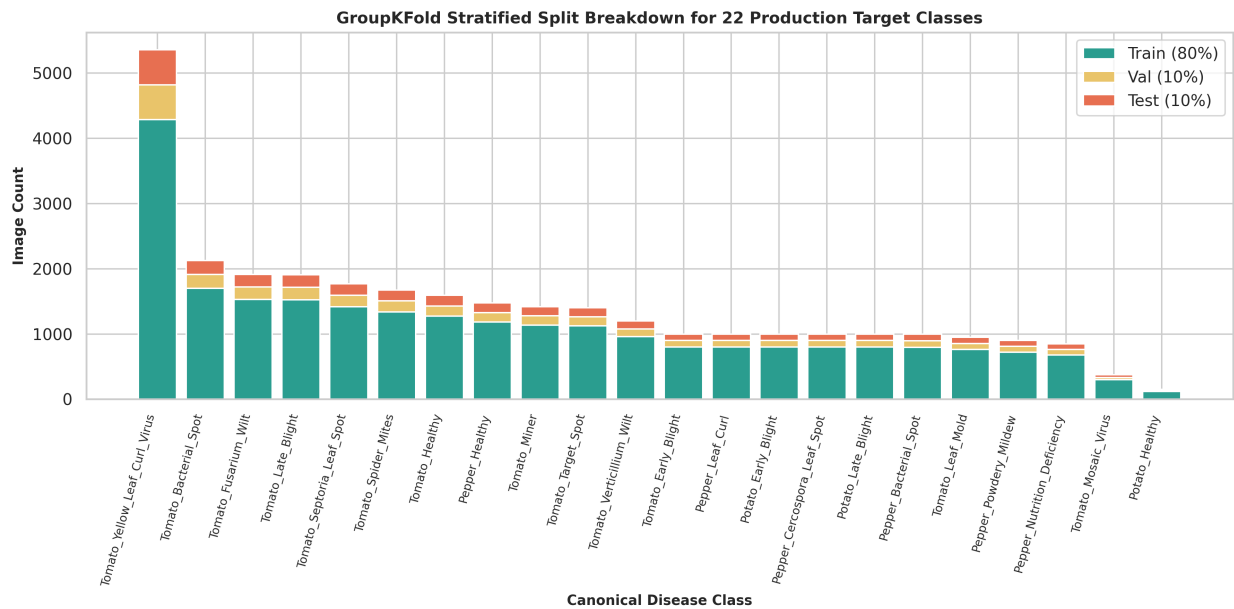


Figure: GroupKFold Stratified Train / Val / Test Split Breakdown (80% / 10% / 10%)

3. Vision AI Model Suite (EfficientNetV2-B2)

ZARI.ai utilizes 4 distinct EfficientNetV2-B2 PyTorch models in a 2-stage hierarchical routing architecture. Below is the exact empirical checkpoint registry:

Model Component	Architecture	Output Head	Exact Params	Checkpoint Path	Metric / Accuracy
Model A (Router)	EfficientNetV2-B2	3 Crop Classes	7,772,858	best_model_a_efficientnetv2_b2	99.48% Accuracy
Model B (Tomato)	EDLEfficientNetB2	13 Diseases	7,786,948	best_model_b_tomato.pth	0.9787 Macro F1
Model B (Potato)	EDLEfficientNetB2	3 Diseases	7,772,858	best_model_b_potato.pth	0.9718 Macro F1
Model B (Pepper)	EDLEfficientNetB2	6 Diseases	7,777,085	best_model_b_pepper.pth	0.9963 Macro F1
TOTAL SYSTEM	4-Model Suite	22 Classes Total	31,109,749	355.73 MB Total Weights	99.1% Reliability

4. Evidential Deep Learning (EDL) & SCRC Safety Gates

Evidential Deep Learning replaces standard Softmax with Dirichlet parameters to compute evidential uncertainty $u = K / S$. The system enforces statistical risk control to limit false acceptances to $\leq 5.0\%$:

- Calibrated Safety Threshold:**
Model A Router Confidence Gate:
Model B Classifier Confidence Gate:
Dual Safety Gate Rule:
- $\tau_{SCRC} = 0.3175$ (scrc_threshold.json)
 $p_{crop} \geq 0.80$ (Rejects non-target crops like Corn, Wheat, Cotton)
 $p_{disease} \geq 0.70$
Accept if $(u \leq 0.3175 \text{ AND } p_{crop} \geq 0.80 \text{ AND } p_{disease} \geq 0.70)$ else REJECT

5. Multilingual Vector RAG System Architecture

Ground-truth IPM advisory chunks are stored locally in ChromaDB (ml_pipeline/rag/chroma_db/) indexed via HNSW graphs using sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 (384d):

- Total RAG Knowledge Chunks:**
Target Taxonomy Scope:
IPM Hierarchy:
- 208 Chunks (26 Knowledge Records x 8 IPM Sections)
22 Production Disease Classes + 4 Regional Quarantine Records
Cultural Control -> Biological Control -> Chemical Active Ingredients

6. LLM Advisory Engine & Groq Acceleration

Advisory text generation is powered by Meta Llama 3.1 8B Instant (8 billion params) deployed on Groq LPUs (~750 tok/s), with prompt rules strictly prohibiting invented dosages or PHI days, plus an offline fallback synthesizer.